

biometryassist: an R package to aid the design and analysis of agronomic experiments

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Summary

We developed the *biometryassist* R package (Nielsen et al. 2026) to provide helper functions and documentation supporting the design and analysis of agronomic field experiments, and have published it on the Comprehensive R Archive Network (CRAN). Its functions cover the common steps of a field trial workflow, including generating and randomising experimental designs, checking fitted mixed models, and presenting the results. Model fitting itself is outside the scope of the package, but *biometryassist* provides tools to support users in assessing whether fitted models are appropriate for their data. We designed the package for agricultural researchers who need robust analyses but may have limited statistical or programming experience.

The package also serves as a companion to our biometry training workshops at the Biometry Hub, Adelaide University, where we use it to reduce the cognitive load on participants so that emphasis remains on statistical concepts and interpretation rather than software mechanics. The functions and documentation deliberately align with the concepts taught, so participants can continue using the package in their own work after the workshops, alongside researchers who adopt it independently. We maintain a companion website with comprehensive documentation and usage examples at <https://biometryhub.github.io/biometryassist/>.

Statement of need

Agricultural researchers are often expected to design and analyse their own experiments, yet rigorous experimental design and mixed-model analysis draw on statistical methods that fall outside most researchers' training (Kozak 2016). In R, these tasks are typically spread across several packages, each with its own conventions and syntax, and learning to navigate them adds effort that has little to do with the statistics. This can distract from understanding the underlying concepts and interpreting the results.

We wrote *biometryassist* in response to this, having seen through a long-running biometry training workshop series that participants consistently struggled with software mechanics as well as the statistics. The package provides a consistent interface across the key steps of an agricultural experiment, allowing users to concentrate on statistical reasoning instead of the software itself. We built it primarily for researchers, agronomists, and applied scientists who require statistically sound analyses but may not have extensive programming experience, though it applies wherever designed experiments are analysed.

37 State of the field

38 Several R packages address individual stages of the field trial process. Experimental designs
39 can be generated by *agricolae* (de Mendiburu 2023), *dae* (Brien 2025), and *edibble* (Tanaka
40 2023); linear mixed models can be fitted with *ASReml-R* (The VSNi Team 2023), *lme4* (Bates
41 et al. 2015), or *nlme* (Pinheiro et al. 2026); and post-hoc inference can be provided by *emmeans*
42 (Lenth and Piaskowski 2026). No single tool spans the full process however, and combining
43 several requires reconciling different syntax, conventions, and documentation standards.

44 We did not build *biometryassist* to replace these packages, but to wrap and extend them within
45 one consistently documented interface. Our case for a new package rather than contributing
46 to an existing one is straightforward: no existing package had the scope required, and the
47 fragmentation is architectural. Improving post-hoc testing in *emmeans*, for example, would not
48 address the design or reporting stages, nor reduce the number of packages a user must load.
49 The documentation in *agricolae* was a particular motivation for us: its sparse and inconsistent
50 help pages were a recurring obstacle in our workshops, and approachable documentation was
51 one of our explicit goals.

52 The package also provides capabilities not available elsewhere. Buffer rows and columns can
53 be added to generated designs for example, which is common practice in agronomic trials but
54 unsupported by other design packages. Designs can also be exported directly to formatted
55 Excel workbooks with a colour-coded field layout together with a spreadsheet experiment field
56 book. The analysis functions reduce a typical analysis from around 50 lines of code across
57 several packages to about 10, using one package, or two with *ASReml-R*. Fitting models
58 remains outside the scope of *biometryassist* as the complexity of this task is better handled
59 by the specialised packages already available. The package is fully open source and released
60 under the MIT License.

61 Software design

62 We organised the functions in *biometryassist* into three categories: design, for generating
63 and randomising experimental layouts (*design()*), adding buffer plots (*add_buffers()*),
64 and exporting them to formatted Excel workbooks (*export_design_to_excel()*); analysis,
65 covering model checking (*resplot()*, *variogram()*), post-hoc testing and prediction
66 (*multiple_comparisons()*, *pairwise_comparisons()*, *reference_comparisons()* and
67 *autoplot()*); and utilities, including installation helpers (*install_asreml()*) and reporting
68 templates (*use_template()*). This structure mirrors the stages of a typical field trial and
69 follows the sequence of topics we cover in the workshops.

70 We chose function names and terminology to reflect how concepts are taught rather than how
71 they are implemented, intending names such as *design()* and *multiple_comparisons()* to
72 be self-explanatory to a researcher with little prior R experience.

73 For model fitting, *biometryassist* supports *ASReml-R* (The VSNi Team 2023), *lme4* (Bates
74 et al. 2015), *nlme* (Pinheiro et al. 2026), and base *aov()*. We recommend *ASReml-R* for the
75 mixed-model analyses of typical agronomic field trials, and concentrate most development effort
76 there, but it is not a hard dependency; the package loads and functions correctly without it, and
77 we have written it to support alternative backends where possible. This keeps *biometryassist*
78 useful across institutional settings where a commercial *ASReml-R* licence may not be available.

79 We designed the package to extend as the R ecosystem evolves. We add support for additional
80 model classes in response to user needs, such as the recent addition of *lme4*, *sommer* and
81 *glmmTMB* support in *resplot()* and the **_comparisons()* functions, and will continue to do
82 so as new packages are developed and gain traction.

Research impact statement

biometryassist began as a collection of analysis scripts we shared with workshop participants, before we formalised it as a package distributed through CRAN. Our workshop series has run for more than eight years, and the package features in each one; we encourage and support participants to apply it in their own work afterwards. Beyond the workshops, researchers have adopted it independently of our team, and it has been downloaded more than 25,000 times from CRAN as of June 2026. It has also been cited in published work spanning agricultural and other applied disciplines (Fanning et al. 2022; Reed-Métayer et al. 2025; Wenham et al. 2026; Kesser et al. 2026), demonstrating use beyond the field of agronomy for which we originally designed it.

AI usage disclosure

We used Anthropic's Claude to suggest edits improving the clarity and flow of text we had drafted, and to assist with some recent refactoring of the package's code. In both cases we reviewed all AI-generated or AI-edited content, and confirmed the correctness of code changes with the package's existing suite of formal unit tests. We take full responsibility for the final content of both the software and this paper.

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